

WELL COMPLETION REPORT (SYNTHETIC)

OIL-BGJ-009 - Baghjan-009

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000105
Field / District	Baghjan / Tinsukia
Block	OALP-IV-03
Basin	Assam-Arakan
Formation	Kopili (KOP)
Spud / Completion	2021-03-10 / 2021-08-04
Measured / TVD	3016 m / 2996 m
Trajectory	Vertical; inc 1.5 deg; az 223.2 deg
Formation Interval	1778-2866 m
Lithology	Compact shale with sandstone streaks
Pore Pressure Reference	4000 psi
Fracture Gradient	15.43 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	High (59.3/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	754	Surface
9 5/8 in casing	1749	Intermediate
7 in liner/casing	2654	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2021-06-10 1987 m	NPT	Critical	Drilling parameters were stable but operations were suspended. Mitigation: Maintain critical spares for recurring equipment failures. Lesson: Recurring equipment downtime should influence planning and spares.
2021-07-12 2586 m	Stuck Pipe	Critical	Hookload became erratic during pipe movement. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Avoid prolonged static periods in high-drag interval.
2021-03-29 533 m	Fishing Operation	High	Downhole tool remained unrecovered after freeing attempts. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2021-05-17 1516 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2021-05-24 1659 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2021-06-24 2245 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Avoid prolonged static periods in high-drag interval.
2021-07-23 2796 m	Torque Spike	High	Rotary speed fluctuated during constant pump rate. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Use offset torque trend to set adaptive operating envelope.

Completion Summary

The synthetic well reached 3016 m MD in the Kopili interval. The NWIS knowledge record links 18 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.